

Progress Report #3

Project Title: Email Spam Detector: Chrome Extension
with Python Backend

Course: 4495 – Applied Research Project, Section: 002

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Work Logs

Date	Hours Worked	Description of Work Done
Jan 14, 2025	2	Conducted initial research on spam detection techniques, including keyword filtering, heuristic algorithms, Bayesian classifiers, and machine learning. Compiled findings into a research document
Jan 16, 2025	1.5	Researched tools and components required for Chrome extension development. Explored Chrome API documentation and feasibility of integrating with Gmail.
Jan 20, 2025	3	Defined project structure, created a preliminary framework for backend and frontend integration, and discussed Flask API implementation. Documented key design decisions.
Jan 22, 2025	0.5	Created project repository, added initial files and organized folder structure for efficient project management. Encountered difficulty in designing an optimal file structure for seamless integration between frontend and backend.
Jan 25, 2025	1.5	Finalized project proposal, incorporated feedback, and documented technical specifications. Submitted proposal for review.
Jan 27,2025	1.5	Research done on the dataset for project.
Jan 30,2025	2	Started working on project using Jupyter, with the help of Python.
Feb 01,2025	1.5	Faced difficulty in pushing folders on GitHub, tried to figure out the problem.
Feb 05,2025	2.5	Worked together as team to find out errors encountered while data cleaning
Feb 07,2025	2	Worked on pushing files using command prompt and visual studio.

Feb 08,2025	3	Created project report, analysed the work done and challenges faced in project.
Feb 09,2025	2.5	Finalised the project report and submitted for grading.
Feb 12,2025	1.5	Installation of libraries in the independent environment
Feb 17,2025	2	Started work on Front end
Feb 19,2025	3.5	Did model training and worked on front end for user using html and javascript
Feb 23,2025	4	Created reoport and made video for submission.
Feb 24, 2025	2	Researched and analyzed techniques for automatic email detection within a Chrome Extension. Explored DOM parsing and Gmail API integration.
Feb 25, 2025	2.5	Reviewed Chrome Extension security policies and identified potential limitations affecting automatic email extraction. Documented initial findings.
Feb 26, 2025	3	Implemented initial email extraction functionality using DOM parsing. Encountered issues with restricted email access due to security policies.
Feb 2, 2025	3	Tested different ways to bypass security restrictions while complying with Chrome policies. Attempted content script injection to read email content.
Feb 28, 2025	2.5	Attempted to integrate Gmail API for automatic email detection. Faced permission constraints and execution limitations. Documented findings.
Mar 1, 2025	2	Worked on refining the API request structure to handle incoming email data efficiently. Implemented request validation checks.
Mar 2, 2025	3	Improved spam classification model by refining preprocessing techniques for better accuracy. Tested different feature extraction methods.

Mar 3, 2025	2.5	Researched alternative classification models for spam detection. Compared traditional ML (Naïve Bayes, SVM) with deep learning-based approaches (LSTM, CNN).
Mar 4, 2025	2	Developed API endpoints to process extracted email content and return classification results. Ensured efficient request handling.
Mar 5, 2025	3	Conducted integration testing between Chrome Extension and API backend. Fixed minor request-response inconsistencies.
Mar 6, 2025	2.5	Debugged CORS issues affecting communication between the Chrome Extension and backend API. Implemented solutions to improve request-response handling.
Mar 7, 2025	3	Explored UI improvements for the extension, including a clear classification indicator for spam vs. non-spam emails. Added initial styling and layout changes.
Mar 8, 2025	3	Enhanced manual email extraction within the extension to improve user experience. Integrated API call functionality for real-time classification.
Mar 9, 2025	2.5	Conducted speed and performance tests on email classification response time. Optimized API query handling to improve response speed.
Mar 10, 2025	4	Implemented logging mechanisms to track API requests and responses for debugging purposes. Conducted initial tests on backend stability.
Mar 12, 2025	3	Conducted end-to-end testing of the extension and backend integration. Validated classification accuracy and frontend response time.
Mar 13, 2025	2.5	Created detailed documentation on the current system design, listing API endpoints, request-response format, and extension functionality.

Mar 14, 2025	4.5	Documented failed attempts at automatic email detection and explored alternative approaches for future implementation. Planned next steps, including possible use of OAuth for accessing Gmail data securely.
March 15,2025	3	Prepared progress report 2
Mar 16, 2025	2.5	Attempted to refine Gmail API authentication using OAuth but faced persistent authorization issues. Explored alternatives, but no breakthrough.
Mar 17, 2025	3	Debugged backend API inconsistencies that affected response accuracy. Minor improvements but no significant progress.
Mar 18, 2025	2.5	Refined spam classification model, tested variations of feature extraction, but accuracy improvements were minimal.
Mar 19, 2025	3	Explored an alternative way to extract emails using Chrome DevTools protocol, but implementation is complex and time-consuming.
Mar 20, 2025	2.5	Conducted further debugging of front-end and backend integration. Persistent issues with handling large email bodies efficiently.
Mar 21, 2025	3	Attempted to optimize UI responsiveness but faced challenges with rendering results dynamically.
Mar 22, 2025	4	Explored spam detection model tuning, including hyperparameter optimization, but with minimal improvements.
Mar 23, 2025	4.5	Revised project documentation to reflect current progress and failures. Tested different debugging strategies for API issues. Created and submitted project report 3

Description of Work Done

In the past few weeks, I focused on overcoming key obstacles in integrating the Chrome Extension with the Python backend. However, progress has been slow due to various technical limitations. The main activities included:

- **Backend Development:**
 - Refined spam classification model but saw only slight improvements.
 - Debugged API inconsistencies and response delays, but certain inefficiencies persist.
 - Explored optimizing request handling for large email bodies but faced memory constraints.
 - **Chrome Extension Integration:**
 - Attempted OAuth-based Gmail access but encountered persistent permission issues.
 - Tried an alternative approach using Chrome DevTools protocol but faced implementation hurdles.
 - UI optimizations were partially successful, but real-time response issues remain.
 - **Testing & Debugging:**
 - Conducted multiple debugging sessions but could not fully resolve key integration issues.
 - Documented unsuccessful attempts and explored alternative strategies for future improvements.
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Current Challenges and Next Steps

- **Challenges Faced:**
 - OAuth authentication remains a significant roadblock.
 - Chrome security policies restrict automated email extraction.
 - Limited improvements in spam classification accuracy despite optimizations.
 - UI responsiveness issues persist due to API latency.

- **Next Steps:**

- Re-evaluate alternative data extraction techniques.
- Further optimize the spam detection model with advanced feature engineering.
- Improve backend request handling for better API efficiency.
- Focus on incremental UI improvements to enhance user experience.

Overall, progress has been slower than expected, with repeated challenges in authentication and performance optimization. Moving forward, the focus will be on resolving key technical bottlenecks to ensure a functional prototype.

Repo Check-In of Implementation:

- Minor updates to app.py for improved request handling.
 - Debugging logs added to background.js and content.js.
 - Documentation updates reflecting integration issues and alternative approaches.
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Conclusion

Despite consistent efforts, the project has faced multiple challenges, including OAuth authentication issues, Chrome security restrictions, and API performance inefficiencies. While some improvements have been made, progress has been slower than anticipated. Moving forward, refining email extraction methods and enhancing model accuracy remain key priorities to achieve a more robust and functional implementation.